Bikal Shrestha

Spatial Data Analyst & Data Scientist

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EDUCATION

MS, Geographic Information Science for Remote Sensing May 2026
Clark University, Worcester, Massachusetts

MS in Data Analytics December 2023
Clark University, Worcester, Massachusetts

BSc (Hons) in Computing 2016 – 2019
Softwarica College of IT and E-commerce, Kathmandu, Nepal (affiliated with Coventry University)

EXPERIENCE

Senior Lecturer June 2026 – Present
Softwarica College of IT and E-commerce, Kathmandu

- Designed and delivered an 11-module lecture series spanning perceptrons and backpropagation through CNNs, transformers, and deep reinforcement learning.
- Authored four hands-on PyTorch labs covering MNIST data pipelines, LSTM forecasting, CIFAR-10 image classification, and a Vision Transformer built from scratch.
- Supervising and assessing individual master's capstone projects applying neural networks to real-world datasets.

Data Analyst / Backend Developer July 2020 – July 2021
Softwarica College of IT and E-commerce, Kathmandu

- Cleaned millions of records across 250 Moodle database tables using SQL and Tableau Prep.
- Built executive, marketing, and performance dashboards in Tableau for institutional reporting.
- Designed and shipped an Android app and REST APIs using Flutter and Node.js.

Teaching Assistant June 2019 – July 2020
Softwarica College of IT and E-commerce, Kathmandu

- Supervised ~40 undergraduate students per semester on independent research projects.
- Mentored teams building hardware/software projects, including an electric vehicle prototype.

Staff Manager January 2017 – September 2019
Hotel Pokhara Peace

- Managed daily operations and staff coordination across departments.
- Developed an internal web presence using HTML, CSS, and JavaScript.

SELECTED RESEARCH PROJECTS

Rasuwa Transboundary Flood Aug 2026

Multi-sensor change detection (Sentinel-1, Sentinel-2) and a terrain-derived flood corridor (SRTM) for the 26 August 2026 Bhote Koshi–Trishuli glacial flood in Rasuwa, Nepal; validated against the Humanitarian OpenStreetMap Team ground survey and published as a public information site.

rasuwaflood.bikal3.com.np

BhumiScan: Earth-Embedding Search for Nepal Jul 2026

Country-scale visual similarity search over 22,676 Clay v1.5 Sentinel-2 embedding cells, with 2020–2025 land-change detection scored by Sentinel-1 radar. 73.7% macro precision@10 against ESA WorldCover labels (5.9× over random); runs entirely client-side.

bhumiscan.bikal3.com.np

Dual-Branch U-Net for Precipitation Downscaling Jan 2026

Dual-branch CNN fusing GOES-17 imagery with SRTM topography to downscale NASA IMERG precipitation from 10 km to 250 m over the Big Island of Hawai'i, using ~165 HCDP rain gauges.

github.com/bikal3/dual-branch-unet-precip

Mapping Invasive Species: Hadwen Arboretum Nov 2025

GIS-based survey of invasive plants across 26 acres in Worcester, MA, finding 42.6% of the arboretum contains at least one invasive species; presented as a narrative web app with density maps, a threat index, and a management effort estimator.

github.com/bikal3/arboretum-invasive-species

MappingAfrica: Satellite Agricultural Field Segmentation May 2025

U-Net semantic segmentation of farmland in Zambia from multi-spectral imagery: 81.79% pixel accuracy and 43.31% mIoU on the MappingAfrica v2.0.0 test split, with in-browser inference.

github.com/bikal3/mappingafrica-unet

Peru Wildfire Patterns in Protected Areas and Indigenous Territories Mar 2025

Spatio-temporal analysis of 32,000+ NASA FIRMS hotspots and MODIS burned area (2000–2024), with vulnerability assessment across protected areas and indigenous territories; delivered as an interactive dashboard.

github.com/bikal3/peru-wildfire

California Wildfire Analysis Nov 2024

Spatiotemporal analysis of 38 years (1984–2022) of MTBS burn extent and severity against ERA5 climate records using geospatial Python; delivered as an interactive dashboard.

github.com/bikal3/mtbs_wildfires

Nepal GLOF Explorer Dec 2023

Glacial lake outburst flood hazard mapping for 25 lakes across the Nepal Himalaya (2000–2024), combining ML-based risk classification with satellite change detection.

github.com/bikal3/himalaya-glof

Coursework projects: machine-learning detection and prediction of urban heat islands; landscape dynamics in North Carolina after Hurricane Helene (2024), including post-hurricane water extent and pre/post NDVI; parcel-level urban heat island analysis in Worcester, Massachusetts.

TECHNICAL SKILLS

Remote sensing	Google Earth Engine, TerrSet; Sentinel-1/2, MODIS, NASA FIRMS, IMERG, GOES, SRTM
GIS	ArcGIS Pro, ArcGIS Online, QGIS, TerrSet
Machine learning	PyTorch; CNNs and U-Net segmentation, LSTMs, Vision Transformers, foundation-model embeddings
Programming	Python, R, SQL, JavaScript, Node.js, Flutter, HTML, CSS
Analytics	Excel, Tableau, Tableau Prep, Power BI

CERTIFICATIONS

Suitability Modeling: Creating a Simple Suitability Model	April 2025
Terrain Analysis Using ArcGIS Pro	March 2025
NASA Open Science 101	August 2024
Google Data Analytics Certificate, Coursera	June 2021